Unit III: Colloidal State of Matter

Concept of colloidal and its types, Different system of colloids, Dispersed phase and dispersion medium, Methods of preparation of colloidal solutions, Dialysis and electro dialysis, Properties of colloidal solution Brownian Movement, Tyndall effect, Electrophoresis and coagulation, Protection and protective colloids.

Emulsion: Types, preparation, properties and uses, Application of colloids chemistry in different industries.

Concept of colloidal and its types

The nature of the solution formed depends upon the size of the solute particles. If the size of the solute particles is less than 1 nm it will form true solution but when the size is between 1 to 1000 nm then it will form colloidal solution. When the size of solute particles is greater than 1000 nm it will form a suspension. Therefore we may conclude that colloidal solution is an intermediate state between true solution and suspension.

S.No.	Name of Property	True Solution	Colloids Solution	Suspension
1.	Size	Size of particles is less than 1 nm	Size of particles is between 1nm and 1000 nm	Size of particles is greater than 1000 nm
2.	Filterability	Pass through ordinary filter paper and also through animal membrane.	Pass through ordinary filter paper but not through animal membrane.	Do not pass through filter paper or animal membrane.
3.	Settling	Particles do not settle down on keeping	Particles do not settle down on their own but can be made to settle down by centrifugation.	Particles settle down on their own under gravity.
4.	Visibility	Particles are invisible to the naked eye as well as under a microscope.	Particles are invisible to the naked eye but their scattering effect can be observed with the help of a microscope.	Particles are visible to the naked eye.
5.	Separation	The solute and solvent cannot be separated by ordinary filteration or by ultra filteration.	The solute and solvent cannot be separated by ordinary filteration but can be separated by ultra-filteration.	The solute and solvent can be separated by ordinary filteration.
6.	Diffusion	Diffuse quickly	Diffuse slowly	Do not diffuse

Phases of Colloids Solution

Colloids solutions are heterogenous in nature and always consist of at least two phases: the dispersed phase and the dispersion medium.

- **Dispersed Phase:** It is the substance present in small proportion and consists of particles of colloids size (1 to 100 nm). Such particles are normally easily visible in an optical microscope.
- **Dispersion Medium:** It is the medium in which the colloids particles are dispersed.

The colloidal state is very widespread in biological systems and in many practical situations. Unlike solutions, colloids do not constitute a solute dissolved in the solvent phase. Rather, the solute phase is dispersed in the solvent phase. For example, in a colloidal solution of sulphur in water, sulphur particles constitute the 'dispersed phase' and water is the 'dispersion medium'. Each of the two phases namely, dispersed phase and dispersion medium can be solid, liquid or gas. Thus, different types of colloidal solutions are possible depending upon the physical state of the two phases. For example:

Sol – It is a suspension of minute solid particles in a liquid.

Emulsion – It is a colloid between two or more liquid with one consisting a dispersion of another liquid.

Foam – It consists of gas dispersed in solid or liquid.

Aerosol – It consists of minute liquid or solid particles in a gas.

Different types of colloidal solutions and their examples are shown in Table:

Dispersed phase	Dispersion medium	Name	Examples
Solid	Solid	Solid-sol	Ruby glass, alloys
Solid	Liquid	Sol	paint, ink
Solid	Gas	Aerosol	Smoke, haze
Liquid	Solid	Gel	Curd, cheese
Liquid	Liquid	Emulsion	Milk, Cream
Liquid	Gas	Liquid Aerosol	Cloud, mist
Gas	Solid	Solid form	Cork, pumice stone
Gas	Liquid	Foam	Froths of air

Note:

- Gases cannot form a colloidal solution between themselves, because they form homogenous mixtures.

- Out of the various types of colloidal solutions listed above, the most common are sols (solid in liquid type), gels (liquid in solid type) and emulsions (liquid in liquid type). If the dispersion medium is water then the 'sol' is called a hydrosol; and if the dispersion medium is alcohol then the 'sol' is called an alcosol.

Classification

Depending upon the interaction between dispersed phase and the dispersion medium colloidal solutions have been classified into two categories.

1. **Lyophilic colloids:** The word Lyophilic means solvent lover. Lyophilic colloidal solutions are those in which the dispersed phase has a great affinity (or love) for the dispersion medium. Substances like gum, gelatine, starch etc. when mixed with suitable dispersion medium, directly pass into colloidal state and form colloidal solution. Therefore, such solutions are easily formed simply by bringing dispersed phase and dispersion medium in direct contact with each other. However, these colloidal solutions have an important property i.e. they are **reversible** in nature. This means that once lyophilic colloidal solution has been formed then dispersed phase and dispersion medium can be separated easily. Once separated these can again be formed by remixing the two phases. These sols are quite stable. If water is used as dispersion medium then it is termed as hydrophilic colloid.
2. **Lyophobic Colloids:** The word Lyophobic means solvent hating. Lyophobic colloidal solutions are those in which the dispersed phase has no affinity for the dispersion medium. Metals like Au, Ag and their hydroxides or sulphides etc., when simply mixed with dispersion medium do not pass directly into colloidal state. These sols have to be prepared by special methods. These sols can be readily precipitated and once precipitated they have little tendency to go back into the colloidal state. Thus these sols are **irreversible** in nature. Also they are not very stable and require a stabilizing agent to remain in the colloidal form. In case water is used as dispersion medium it is called as hydrophobic sol.

Methods of preparation of colloidal solutions

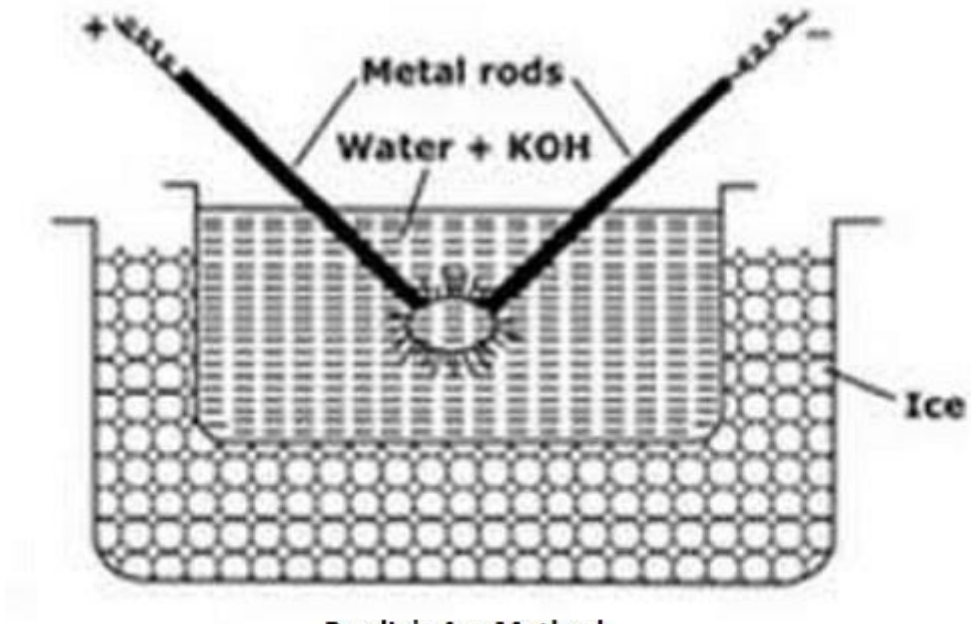
The lyophilic sols can be prepared directly by mixing the dispersed phase with the dispersion medium. For example, colloidal solutions of starch, gelatin, gum etc. are prepared by simply dissolving these substances in hot water. Similarly, a colloidal sol of cellulose nitrate is obtained by dissolving it in alcohol. The resulting solution is called collodion.

However, lyophobic colloids cannot be prepared by direct method. Hence two types of methods are used for preparing lyophobic colloids. These are:

- Physical methods
- Chemical methods

Physical methods

- **Bredig's Arc Method:** These methods are employed for obtaining colloidal solutions of metals like gold, silver, platinum etc.**.** An electric arc is struck between the two metallic electrodes placed in a container of water. The intense heat of the arc converts the metal into vapours, which are condensed immediately in the cold water bath. This results in the formation of particles of colloidal size. We call it as gold sol.



Bredig's Arc Method

- **Peptisation:** Peptisation is the process of converting a freshly prepared precipitate into colloidal form by the addition of a suitable electrolyte. The electrolyte is called peptising agent.
  For example when ferric chloride is added to a precipitate of ferric hydroxide, ferric hydroxide gets converted into reddish brown coloured colloidal solution. This is due to preferential adsorption of cations of the electrolyte by the precipitate. When $FeCl_3$ is added to $Fe(OH)_3$, Fe^{3+} ions from $FeCl_3$ are adsorbed by $Fe(OH)_3$ particles. Thus the $Fe(OH)_3$ particles acquire + ve charge and they start repelling each other forming a colloidal solution.

Chemical Methods: By oxidation

- Sulphur sol is obtained by bubbling H_2S gas through the solution of an oxidizing agent like HNO_3 or Br_2 water, etc. according to the following equation **:**

$$Br_2 + H_2S \rightarrow S + 2\,HBr$$

$$2\,HNO_3 + H_2S \rightarrow 2\,H_2O + 2\,NO_2 + S$$

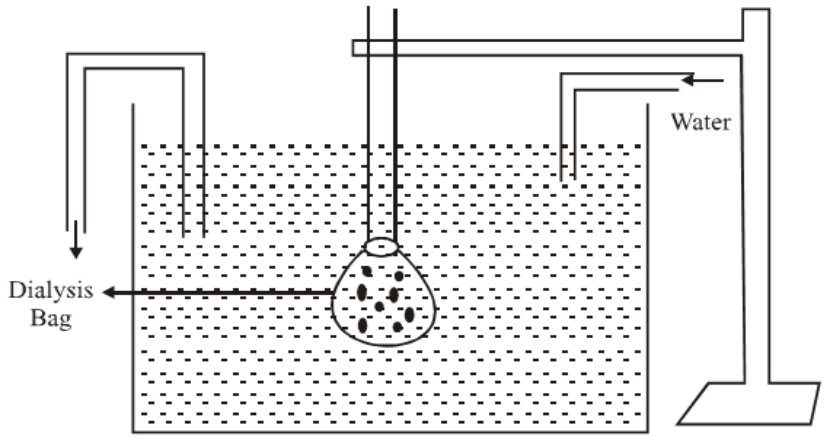
$Fe(OH)_3$ sol, As_2S_3 sol can also be prepared by chemical methods.

Purification of Colloidal Solution

When a colloidal solution is prepared it contains certain impurities. These impurities are mainly electrolytic in nature and they tend to destabilise the colloidal solutions. Therefore colloidal solutions are purified by the following methods:

- Dialysis
- Electrodialysis

Dialysis: The process of dialysis is based on the fact that colloidal particles cannot pass through parchment or celloplane membrane while the ions of the electrolyte can. The colloidal solution is taken in a bag of cellophane which is suspended in a tub full of fresh water. The impurities diffuse out leaving pure colloidal solution in the bag. This process of separating the particles of colloids from impurities by means of diffusion through a suitable membrane is called dialysis.



Dialysis

Electrodialysis: The dialysis process is slow and to speed up its rate, it is carried out in the presence of an electrical field. When the electric field is applied through the electrodes, the ions of the electrolyte present as impurity diffuse towards oppositely charged electrodes at a fast rate. The dialysis carried out in the presence of electric field is known as electrodialysis.